

Articles

UK collaborative randomised trial of neonatal extracorporeal membrane oxygenation

UK Collaborative ECMO Trial Group*

Summary

Background Extracorporeal membrane oxygenation (ECMO) is a complex and expensive technology that can be used to provide temporary support during respiratory failure. Its value for mature newborn infants is controversial because of varying interpretation of the available evidence. We undertook a collaborative trial throughout the UK to assess whether a policy of referral for ECMO has a beneficial effect on survival to 1 year without severe disability in comparison with conventional management.

Methods Between 1993 and 1995, 185 mature (gestational age at birth ≥ 35 weeks, birthweight ≥ 2 kg) newborn infants with severe respiratory failure (oxygenation index ≥ 40) were enrolled from 55 hospitals in a randomised comparison of either referral to one of five specialist centres for consideration of ECMO or continued intensive conventional management at the original hospital. The most common diagnoses were persistent pulmonary hypertension due to meconium aspiration, congenital diaphragmatic hernia, isolated persistent fetal circulation, sepsis, and idiopathic respiratory distress syndrome. Of the infants allocated ECMO, 84% received this support. Recruitment to the trial was stopped early (November, 1995) by the trial steering committee on the advice of the independent data-monitoring committee, because the data accumulated showed a clear advantage with ECMO. 124 children were enrolled before December, 1994; those who survived to 1 year of age have undergone neurological assessment at that age (by one of three developmental paediatricians unaware of treatment allocation).

Findings Overall, 81 (44%) infants died before leaving hospital, and two are known to have died later. Death rates differed between the two trial groups; 30 of 93 infants allocated ECMO died compared with 54 of 92 allocated conventional care. The relative risk was 0.55 (95% CI 0.39–0.77; $p=0.0005$), which is equivalent to one extra survivor for every three to four infants allocated ECMO. The difference in survival applied irrespective of the primary diagnosis, disease severity, and type of referral centre. The benefit of ECMO was also found for the primary outcome measure of death or disability at 1 year (among 124 children enrolled before December, 1994). One child in each group has severe disability (overall Griffiths' developmental quotient <50 , or untestable), and 16 (ten ECMO, six

conventional management) have impairments with a lesser degree of disability.

Interpretation These preliminary results demonstrate the clinical effectiveness of a well-staffed and organised neonatal ECMO service. ECMO support should be actively considered for neonates with severe but potentially reversible respiratory failure.

Lancet 1996; **348**: 75–82

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Introduction

Extracorporeal membrane oxygenation (ECMO) is a complex technique for providing life support in respiratory failure. This technique oxygenates blood outside the body, obviating the need for gas exchange in the lungs, and, if necessary, provides cardiovascular support. ECMO is most commonly used to support mature newborn infants. This application is controversial, principally because of varying interpretation of the available evidence.¹ Most claims about the value of ECMO in comparison with less invasive forms of life support have been based on case series from the Extracorporeal Life Support Organisation² and on studies with historical controls. Such studies have suggested large reductions in mortality but they were conducted at a time when neonatal death rates were falling.^{3,4} The three small randomised controlled trials of neonatal ECMO that have been reported suggested reductions in mortality but were not conclusive.^{5–8} The first two^{5,6} used adaptive designs, which may have introduced bias if the threshold for ECMO became more lenient over time, leading to the inclusion of very few infants managed with conventional ventilation. Their results are therefore hard to interpret. The third trial used a standard randomised controlled design, but the number of infants recruited was small and the study has been reported only as conference abstracts.^{7,8}

Neonatal ECMO was introduced into the UK in 1989. Some clinicians were reluctant to refer potentially suitable infants for ECMO because of concerns that improved survival with the technique might be offset by an increased risk of long-term disability. Others were concerned about the costs of ECMO while uncertainty about its clinical effectiveness and cost-effectiveness persisted.

Against this background, British clinicians agreed to limit neonatal ECMO to use within a randomised controlled trial. This national study compared a policy of transfer for consideration of ECMO with continued conventional management without transfer. Because of concern about quality of survival, the principal outcome measure was death or severe disability at 1 year of age. Our hypothesis was that a policy of referral for ECMO would reduce the proportion of infants who died or

*Study organisation and participants listed at end of article

Correspondence to: Dr D J Field, Department of Child Health, University of Leicester, Clinical Sciences Building, Leicester Royal Infirmary, Leicester LE1 5WW, UK

survived with severe disability in comparison with conventional management. This report concentrates on the findings to hospital discharge, with preliminary information from the 1-year follow-up.

Methods

Trial organisation

The trial was overseen by a multidisciplinary steering group supported by advisory groups on technical aspects of ECMO, developmental follow-up, health economics, pathology, respiratory follow-up, and liaison with parents and relevant voluntary organisations. The clinical coordinating centre was in Leicester and the data-coordinating centre in Oxford. An independent data-monitoring committee was established to review accumulating data; its terms of reference stated that interim results should not be revealed to the steering group unless a very strong reason to alter the protocol or to stop the trial emerged, such as difference between the randomised groups of more than 3 SD. Local research ethics committee approval was obtained for all participating centres.

ECMO support was provided in five centres, four from the start of the trial and the fifth after 40% of the infants had been recruited. All met prespecified criteria. These included 24 h availability of: intensive care facilities; support from staff able to perform cardiopulmonary perfusion, intensive care nurses, neonatologists, and paediatric cardiologists; facilities for cardiopulmonary assessment, echocardiography, and cranial ultrasound; a suitable retrieval transport system; and provision of accommodation and psychosocial support for parents.

The centres where infants were recruited provided conventional management when this was the randomly allocated treatment option. The centres were therefore limited to those accustomed to providing long-term ventilatory support for newborn infants under the care of a paediatrician specialising in neonatology, with 24 h availability of facilities for invasive blood-pressure monitoring and cranial ultrasound. They also had to have provision for psychosocial support for parents of infants in the trial. 83 hospitals meeting these criteria agreed to participate, and infants were recruited from 55 of them.

The criteria for eligibility of infants were: severe respiratory failure (defined as oxygenation index⁹ ≥ 40 , arterial partial pressure of carbon dioxide [PaCO_2] > 12 kPa for at least 3 h); gestational age at birth 35 completed weeks or more and birthweight 2 kg or more; less than 10 days of high-pressure ventilation; age less than 28 days; no contraindication for ECMO support recognised at trial entry (such as ventricular haemorrhage extending beyond the subependymal or choroidal region, irreversible cardiopulmonary disease, a period of asystole with cardiopulmonary resuscitation other than immediately after delivery, or proven necrotising enterocolitis); and no reason to question continuation of conventional management (for example, a major congenital or chromosomal anomaly, or severe encephalopathy). The commonest diagnoses were expected to be persistent pulmonary hypertension of the newborn due to meconium aspiration, congenital diaphragmatic hernia, isolated persistent fetal circulation, sepsis, and idiopathic respiratory distress syndrome.

Trial entry

Infants were considered potentially eligible when the oxygenation index reached at least 35. They were first registered by a telephone call to the clinical coordinating centre. The availability of a cot at an appropriate ECMO centre was then confirmed. Only at that stage was the trial discussed with the infant's parents, aided by written information¹⁰ in an appropriate language, and prepared in consultation with representatives of voluntary organisations concerned with maternity and neonatal issues. If the parents agreed to trial entry, and the infant met the trial entry criteria, eligibility was confirmed by a second telephone call, and the infant was randomly allocated treatment by means of a computerised minimisation algorithm to ensure balance on key

prognostic variables—primary diagnosis, disease severity, referral centre, and the ECMO centre in which a cot was reserved. Allocation was to either referral for consideration of ECMO or conventional management. The clinical coordinating centre then informed the clinician in the referral hospital of the allocation, and contacted the ECMO centre, either to alert the transport team to collect the infant or to release the cot. The clinician in the referring hospital then talked with the parents about the allocation, gave further written information, and arranged their transport to the ECMO centre, if appropriate, with an offer to meet these travel costs.

ECMO treatment schedule

Transport of the infant to an ECMO centre was part of the ECMO policy. Transport was undertaken by a team from the ECMO centre in most cases, by fixed-wing aircraft, helicopter, or road ambulance. ECMO support was withheld if the infant's condition was judged to make ECMO inappropriate (such as a major deterioration or improvement, or a previously unrecognised factor such as a cardiac anomaly). Otherwise, the infant was cannulated and ECMO was started as soon as possible.

The ECMO centres followed an agreed protocol. Both venoarterial (VA) and venovenous (VV) ECMO were permitted, according to the infant's needs and the local surgeon's preference. Infants were cannulated via the right internal jugular vein (VA and VV) and the right carotid artery (VA). All centres used a roller-pump ECMO system, £ in (6.25 mm) tubing circuits, spiral-wound silicone-membrane oxygenators, and in-line counter-current heat exchangers. Once full circuit flow at about $100 \text{ mL kg}^{-1} \text{ min}^{-1}$ for VA and $120 \text{ mL kg}^{-1} \text{ min}^{-1}$ for VV ECMO had been established, the fractional inspired oxygen concentration delivered to the lungs was reduced to 0.21–0.30, and ventilator settings reduced to lung-rest (peak inspiratory pressure < 25 cm water delivered at 10–15 breaths per min; positive end-expiratory pressures 5–10 cm water). Anticoagulation in the extracorporeal circuit was maintained by a continuous infusion of heparin, titrated to achieve an activated clotting time of 160–200 s. Careful attention was paid to maintenance or support of other organ systems, with particular emphasis on fluid and electrolyte balance, nutrition, and renal support. Renal replacement therapy was provided where necessary by haemofiltration of blood from the ECMO circuit. Regular cranial ultrasound examinations were made throughout the ECMO support. Cardiac echocardiography was repeated as appropriate.

After clinical and radiological evidence of lung improvement, ECMO support was reduced and then discontinued with the aid of moderate levels of conventional ventilation (fractional inspired oxygen < 0.5 , peak inspiratory pressure < 30 cm water), and the cannulae were surgically removed. The carotid artery was repaired in some cases, according to clinical findings and surgical preference. After a short period of stabilisation, most infants were transferred back to the original hospital, as soon as a bed was available.

Conventional management schedule

Intensive conventional care was provided in the referring hospitals, based on trial guidelines for continued ventilation. These guidelines recommended liberal use of oxygen, correction of acidosis, maintenance of adequate blood pressure, paralysis, and use of any available pulmonary vasodilator (including nitric oxide) if the infant showed a poor response. Arterial access for blood-pressure monitoring was recommended. Use of high-frequency ventilation and surfactant was allowed.

Outcome measures

The principal endpoint was death or severe disability as assessed at 1 year of age. Outcome was first described at discharge from hospital (vital status, clinical management, and length of hospital stay). The parents of infants who died were asked for consent for necropsy. If consent was given, the examination was carried out

	All infants randomised		Infants enrolled by Dec 1, 1994	
	ECMO (n=93)	CM (n=92)	ECMO (n=63)	CM (n=61)
Primary diagnosis*				
Congenital diaphragmatic hernia	18 (19%)	17 (18%)	11 (17%)	11 (18%)
Isolated persistent fetal circulation	15 (16%)	16 (17%)	11 (17%)	10 (16%)
Persistent pulmonary hypertension due to:				
Meconium aspiration	32 (34%)	37 (40%)	22 (35%)	28 (46%)
Sepsis	14 (15%)	5 (5%)	11 (17%)	3 (5%)
IRDS	8 (9%)	12 (13%)	4 (6%)	7 (11%)
Other	3 (3%)	4 (4%)	2 (3%)	1 (2%)
Other	3 (3%)	1 (1%)	2 (3%)	1 (2%)
Entry criteria*				
Oxygenation index 40–59	55 (59%)	55 (60%)	37 (59%)	36 (59%)
Oxygenation index ≥ 60	37 (40%)	37 (40%)	25 (40%)	25 (41%)
PaCO ₂ >12 kPa for >3 h	1 (1%)	0	1 (2%)	0
Age at entry				
0 days	37 (40%)	42 (46%)	23 (37%)	26 (43%)
1 day	31 (33%)	20 (22%)	23 (37%)	15 (25%)
2–27 days	25 (27%)	30 (33%)	17 (27%)	20 (33%)
Sex				
Female	39 (42%)	34 (37%)	26 (41%)	23 (38%)
Male	54 (58%)	58 (63%)	37 (59%)	38 (62%)
Median (range) gestational age at birth in completed weeks	39 (37–40)	40 (38–41)	39 (37–41)	40 (37–41)
Mean (SD) birthweight in g	3261 (516)	3346 (593)	3241 (516)	3290 (576)
High-pressure ventilation				
0 days	45 (48%)	46 (50%)	28 (44%)	28 (46%)
1 day	27 (29%)	20 (22%)	19 (30%)	17 (28%)
2–9 days	21 (23%)	26 (28%)	16 (25%)	16 (26%)
Use of				
Any pulmonary vasodilators	79 (85%)	78 (85%)	54 (86%)	51 (84%)
Nitric oxide	13 (14%)	11 (12%)	5 (8%)	3 (5%)
Inotropes	75 (81%)	71 (77%)	50 (79%)	45 (74%)
High-frequency ventilation	13 (14%)	14 (15%)	4 (6%)	6 (10%)
Muscle relaxants	81 (87%)	81 (88%)	54 (86%)	57 (93%)
Surfactant	28 (30%)	37 (40%)	16 (25%)	23 (38%)
ECMO centre (in which cot reserved)*				
Glenfield Hospital, Leicester	29 (31%)	35 (38%)	20 (32%)	23 (38%)
Great Ormond Street, London	30 (32%)	26 (28%)	21 (33%)	18 (30%)
Freeman Hospital, Newcastle upon Tyne	14 (15%)	11 (12%)	8 (13%)	7 (11%)
Royal Hospital for Sick Children, Glasgow	17 (18%)	17 (18%)	13 (21%)	11 (18%)
King's College Hospital, London	3 (3%)	3 (3%)	1 (2%)	2 (3%)

*Minimisation factors: primary diagnosis, entry criteria, referral centre, ECMO centre. CM=conventional management; IRDS=idiopathic respiratory distress syndrome. Data are number (%) in allocated groups, unless indicated otherwise.

Table 1: Description of groups at trial entry

under guidelines issued for the Confidential Enquiry into Stillbirths and Infant Deaths,¹¹ with more detailed examination of samples from the organs most likely to be affected by the initial disease and its treatment—the lungs, heart, and brain. These findings will be reported elsewhere.

Telephone contact with parents of surviving infants was initiated by the trial coordinator at the time of discharge home. After correspondence with the infant's general practitioner and health visitor, telephone contact with parents was repeated when the infant was aged 4, 8, and 11 months. At the 8-month and 11-month follow-up, the health visitor completed a short postal questionnaire about the infant's condition and relevant changes in family circumstances. Parents also completed a postal questionnaire about the infant's condition at about 11 months.

When surviving infants were at least 4 months old and at home, parents were asked to complete a questionnaire on their views about participating in the trial. Parents of infants who died were offered the possibility of further contact with trial personnel (including completing the questionnaire), if they so wished.

A standard neurodevelopmental assessment was carried out by one of three developmental paediatricians when surviving infants were 1 year old. This assessment was performed in the infants' home without knowledge of the random allocation. The parents were asked not to comment on the neonatal treatments, and the infant wore a high-necked tunic to hide any scars from ECMO cannulation. The neurological status of the infant was defined in terms of the presence or absence of neuromotor, sensory, and developmental impairment.¹² Neuromotor impairment was

defined as the presence of abnormalities of tone or posture. Infants on regular anticonvulsant treatment and those requiring supplemental oxygen or tube-feeding at the age of 1 year were included in the disabled group.¹³ An infant was classified as severely disabled if the overall Griffiths' developmental quotient was less than 50, or if Griffiths' testing was not possible because the child was blind or had too low a level of functioning. Information was collected about readmissions to hospital and other uses of health service resources. A detailed respiratory assessment at one of two hospitals was also offered (these findings will be reported elsewhere). All surviving infants were flagged on the National Health Service Central Register to facilitate subsequent follow-up.

Economic evaluation

The methods of economic evaluation have been described in detail elsewhere.¹⁴ In summary, information about the costs of providing care was collected from the ECMO centres. The conventional management arm of the trial was costed on data from a detailed study of the costs of neonatal intensive care units;¹⁵ an adjustment factor to take into account the high level of dependency was estimated from a case-note analysis based on infants in the trial.¹⁶ Other costs taken into account included transportation, the initial hospital stay, health service contacts between discharge and 1 year of age, and parental expenses. These data will be reported as part of the final cost-effectiveness analysis of the whole trial. In this paper, information about the charges for the two types of care are discussed.

	All infants randomised		Infants enrolled by Dec 1, 1994	
	ECMO (n=93)	CM (n=92)	ECMO (n=63)	CM (n=61)
ECMO support given	78 (84%)	1 (1%)	53 (84%)	1 (2%)
ECMO support not given				
Too well	3 (3%)	..	2 (3%)	..
Too ill	2 (2%)	..	1 (2%)	..
Died before transfer	5 (5%)	..	4 (6%)	..
Congenital heart disease	4 (5%)	..	3 (5%)	..
Failed cannulation	1 (1%)	..	0	..
Mode of cannulation				
VA	50 (54%)	..	37 (59%)	..
VV	23 (25%)	1 (1%)	14 (22%)	1 (2%)
VA and VV	5 (5%)	..	2 (3%)	..
Not cannulated	15 (16%)	91 (99%)	10 (16%)	60 (98%)
Other management				
Any pulmonary vasodilator	77 (83%)	88 (96%)	50 (79%)	59 (97%)
Nitric oxide	19 (20%)	20 (22%)	7 (11%)	9 (15%)
High-frequency ventilation	10 (11%)	30 (33%)	2 (3%)	15 (25%)
Surfactant	31 (33%)	38 (41%)	14 (22%)	22 (36%)

Table 2: **Actual management between trial entry and discharge from hospital (or death before discharge)**

Statistical analysis

There were no precise estimates of the expected frequency of the principal measure of outcome (death or survival with severe disability) in the conventionally managed group available before the trial. The mortality rate was assumed to be between about 40% and 60%, and the proportion of severe disability in survivors about 5% in both trial groups. An absolute benefit of 20% or more was judged clinically important. After variation of the assumptions about statistical power and significance levels, a sample size of 300 was judged to be sufficient, with the proviso that further guidance from the independent data-monitoring committee would be needed if these assumptions proved incorrect.

The primary analyses are based on the groups as randomly allocated, with secondary analyses stratified by the factors used for minimisation.

Differences between the randomised groups are presented as relative risks and absolute percentage differences or as differences between means or medians, with 95% CI. Statistical significance was tested by χ^2 tests, Fisher's exact tests, *t* tests, and median tests, as appropriate.

Trial recruitment began on Jan 1, 1993. The independent data-monitoring committee met five times to look at the accumulating data. Although information about mortality became available to this committee at early (confidential) meetings, there were few data about severe disability at 1 year of age until 1995.

On Nov 13, 1995, the fifth report was presented to the data-monitoring committee, based on analyses of data describing outcome up to hospital discharge for 180 infants recruited up to Oct 31, 1995, and including 1-year follow-up information about survivors among the 118 infants entered up to Oct 31, 1994. The data-monitoring committee recommended that these data be shown to the trial steering group. The steering group met 10 days later, decided that the trial provided a clear answer, and halted recruitment on Nov 23, 1995.

Outcome	All infants randomised				Infants enrolled by Dec 1, 1994			
	ECMO (n=93)	CM (n=92)	Relative risk of outcome (95% CI)	% absolute reduction (95% CI)	ECMO (n=63)	CM (n=61)	Relative risk of outcome (95% CI)	% absolute reduction (95% CI)
Death before discharge	28 (30%)	54 (59%)	0.51 (0.36–0.73)	29 (15–42)	18 (29%)	37 (61%)	0.47 (0.30–0.73)	32 (15–49)
Known death before age 1 year	30 (32%)	54 (59%)	0.55 (0.39–0.77)	26 (13–40)	20 (32%)	37 (61%)	0.52 (0.35–0.79)	29 (12–46)
Death or severe disability at 1 year	NA	NA	..	..	21 (33%)	38 (62%)	0.54 (0.36–0.80)	29 (12–46)

NA=not available.

Table 3: **Principal measures of outcome**

Results

185 infants were enrolled in the trial (93 allocated ECMO and 92 allocated conventional management). A further 48 infants were registered but not formally enrolled because of death (14), improvement (19, of whom four later died) and parental refusal (15, of whom nine later died). This report describes outcome up to hospital discharge for all 185 infants, with further information about the status at 1 year for the 124 infants who were enrolled before December, 1994.

At trial entry (table 1) about 19% of the infants had congenital diaphragmatic hernia, and 37% had persistent pulmonary hypertension due to meconium aspiration. About 40% had an oxygenation index of 60 or more at trial entry, indicating very severe respiratory disease. Most infants were recruited within a few days of birth, and were mature neonates as judged by gestational age and birthweight. The trial groups were similar in these respects. Before trial entry, pulmonary vasodilators (tolazoline or epoprostenol), inotropes, and muscle relaxants had all been used for about 80% of the infants. Some newer treatments were also recorded but were less commonly used (surfactant [35%], high-frequency ventilation [15%], and nitric oxide [13%]).

Of the 93 infants allocated ECMO, 78 (84%) actually received ECMO support, in most with VA cannulation (table 2). The commonest reason that ECMO was not given was deterioration of the infant, before or after transport. All seven of these infants died. In addition, four infants found on further assessment to have congenital heart disease did not receive ECMO, and two of these later died. One infant died after a failed attempt at cannulation. No infant died during transport.

One infant allocated conventional care received ECMO owing to a series of errors that led to a second randomisation, this time to ECMO. Although this infant received ECMO support (and died at the age of 2 months of respiratory distress syndrome secondary to surfactant protein B deficiency), the data have been analysed within the conventional management group by intention-to-treat.

After trial entry, fewer infants allocated ECMO received pulmonary vasodilators, high-frequency ventilation, and exogenous surfactant.

The principal measures of outcome are shown in table 3. Among all the infants enrolled, 82 (44%) died before discharge from hospital and two are known to have died later, before the age of 1 year. The overall known mortality rate differed significantly between the ECMO and conventional management groups (32 vs 59%, $p=0.0005$), which is equivalent to one extra survivor for every three to four infants allocated ECMO. This benefit of ECMO was sustained when severe disability at 1 year of age was taken into account (table 3; $p=0.002$). Of the infants assessed at 1 year of age, one in each group is severely disabled. Most of the deaths occurred in the neonatal period; the median survival time for those who died was 14 days in the

Outcome	ECMO (n=63)	CM (n=61)
Death		
Before discharge home	18 (29%)	37 (61%)
After discharge home and before age 1 year	2 (3%)	0
Severe impairment and disability	1 (2%)	1 (2%)
Signs of impairment		
Disability not classified as severe	8 (13%)	2 (3%)
Without disability	2 (3%)	4 (7%)
Number of infants with no signs of impairment or disability	32 (51%)	17 (28%)

Table 4: Overall status at 1-year follow-up of 124 infants enrolled by Dec 1, 1994

ECMO group compared with 2 days in the conventional management group ($p=0.04$).

103 infants were discharged home (65 ECMO group, 38 conventional management group). The hospital stay was slightly shorter in the ECMO than the conventional management group but the difference was not statistically significant (median 27 vs 33 days, $p=0.21$).

Outcome at the age of 1 year is currently known only for the 124 infants recruited before December, 1994. Their characteristics at trial entry (table 1), subsequent management (table 2), and mortality differences (table 3) were similar to those of the total trial cohort. Their 1-year status is summarised in table 4. Neither infant who died after discharge from hospital was ever well. One had neurological abnormalities, persistently required supplemental oxygen and tube-feeding, and died of candidal meningitis. The other had total anomalous pulmonary venous drainage and pulmonary vein stenosis, and died as a result of inoperable recurrent pulmonary vein obstruction.

All 67 survivors enrolled before December, 1994, have received a 1-year follow-up visit (43 ECMO group, 24 conventional management group), although one assessment was carried out at 18 months. One of the infants in the ECMO group was too severely disabled to be tested on the Griffiths scale, and one infant in the conventional management group had global delay and was judged severely disabled based on an overall Griffiths' score of less than 50. 16 other infants (ten ECMO group,

six conventional management group) also had signs of impairment. Ten of the 16 had associated disabilities that were not classified as severe (eight ECMO group, two conventional management group). The remaining six infants had an impairment only, without signs of disability. (A full description of these impaired infants will be reported elsewhere.) 49 (40%) of the 124 infants showed no signs of impairment or disability (51% of those allocated ECMO and 28% of those allocated conventional management). The overall rate of impairment with or without disability among survivors is therefore similar in the two trial groups (11/43 [26%] ECMO group vs 7/24 [29%] conventional group).

Lower mortality was seen among infants allocated ECMO in all the prespecified stratified analyses (figure). Among infants with a primary diagnosis of congenital diaphragmatic hernia, the outcome was particularly poor. Of the 18 allocated ECMO, 14 died (one after discharge); all 17 in the conventional group died (relative risk 0.78 [95% CI 0.610–0.996], $p=0.10$). Mortality among infants with a primary diagnosis other than diaphragmatic hernia was 21% in the ECMO group compared with 49% in the conventional management group (relative risk 0.43 [0.27–0.71], $p=0.0006$). As expected, the overall mortality rate was higher in the stratum with more severe respiratory failure at trial entry (oxygenation index ≥ 60), but the rate was lower in the ECMO group in both strata of oxygenation index. The mortality rate was also lower in the ECMO arm whether the referral centre (and hence the site of conventional care) was a teaching hospital or non-teaching hospital.

Discussion

The trial results leave little doubt that allocation to the ECMO policy reduced the risk of death or severe disability. Potential bias was kept to a minimum by strict random allocation, complete follow-up, and assessment of outcome without knowledge of treatment allocation. Estimates of improved survival were sufficiently large and sufficiently precise to rule out an absence of effect, even in the subgroup analyses. No other study of ECMO has these strengths. There are, nevertheless, other issues that need to be taken into account in the interpretation.

The observed differences could be overestimates if they reflect greater enthusiasm within the ECMO teams because of the new technology, or greater experience of providing intensive care for such severely ill infants, rather than an effect of ECMO itself. To guard against this potential bias, hospitals providing conventional management in the trial had to meet criteria indicating substantial experience in providing neonatal ventilatory care. Furthermore, secondary analyses stratified by type of referral hospital (teaching or non-teaching) did not suggest any extra effect depending on the site of conventional care (figure). A stratified analysis by the ECMO centre to which a baby was randomly allocated showed no significant heterogeneity.

Another possible cause of overestimation might be that the conventional management was suboptimal. Although details of management could not be specified in advance, guidelines were provided, and the management actually given was described. As might be expected, more infants in the conventional management group than in the ECMO group were treated with high-frequency ventilation, surfactant, or both (table 2), especially in the

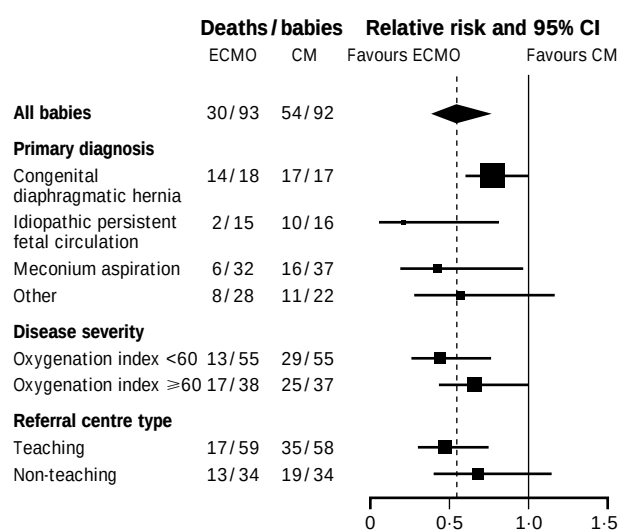


Figure: Effects of ECMO policy on known deaths

Relative risks (squares; proportional to amount of information contributed) and 95% CIs (horizontal lines) are plotted for various subgroups. Overall results shown by the diamond. CM=conventional management.

later years of the study. Whether even greater use of these treatments would have improved outcome after conventional care is uncertain. There is currently little rigorous evidence of their effectiveness for mature newborn infants with severe respiratory failure.

Conversely, the advantages of ECMO may be underestimated. We choose to consider transport as part of the ECMO policy because ECMO is likely to be available in only a limited number of centres. As well as presenting emotional and logistical problems for families, the physical risks of transport and the delayed start to ECMO may mean that the benefits of this technique are greater for infants with easy access to it.¹⁷ There is also concern that use of high-frequency ventilation with or without inhaled nitric oxide may make the transport of such sick infants more difficult. The value of ECMO may also have been underestimated because of the relative inexperience of the staff providing it. Initial analyses do not, however, suggest a learning curve for ECMO, and the trial may have mitigated the effect of inexperience by providing the structure for a standard national service with common training and mutual support among personnel in the ECMO centres.

Another possibility is that ECMO is more useful for some categories of infants than others, and these may have predominated within the trial. For example, the infants registered but not randomised who later died may have deteriorated too rapidly to benefit from ECMO. As the figure shows, however, improved survival with ECMO was seen in all diagnostic categories, even for infants with congenital diaphragmatic hernia; although, as expected from the Extracorporeal Life Support Organisation Registry data,² the overall outcome was disappointing for this group with all of the infants allocated conventional management dying. Management of these patients in most UK centres involves initial stabilisation and delayed surgery. Some centres require evidence of adequate lung function before surgery is undertaken. In the presence of severe pulmonary hypertension, adequate lung function may not be achieved with conventional management alone. There seems to be a strong case, therefore, for further study of the role of ECMO and other treatments (such as nitric oxide, high-frequency ventilation, and surfactant) in the management of all infants with congenital diaphragmatic hernia who require anything more than modest respiratory support (oxygenation index >20) in the first 12 h of life.

The data currently available on long-term morbidity are limited because not all survivors have been reviewed at 1 year. Also, assessment at this age will not detect abnormalities in all areas of development, especially cognitive defects. The 1-year follow-up will be completed in late 1996, and periodic review of all children will continue at least to school age. The preliminary data reported here suggest that the rate of impairment among survivors is similar in the two groups, so the actual number of impaired survivors may be greater after ECMO because of lowered mortality. Nevertheless, even if all the surviving ECMO-allocated infants recruited after November, 1994, prove to be disabled (which is not plausible on the basis of the results so far), this disadvantage would not outweigh the observed survival benefits associated with ECMO.

If ECMO does offer improved survival for mature infants with severe respiratory failure that is not offset by additional risk of disability, what are the implications for

the health services? The issue of cost-effectiveness remains unresolved for the present. An initial cost-effectiveness analysis based on evidence from a systematic literature review was carried out before the trial started.¹⁸ It concluded that an additional life would be gained at the cost of £180 000 (at 1994 prices), but the analysis took account of time to hospital discharge only, and there is much uncertainty attached to the estimate. The mean charge per infant for ECMO was estimated as £17154; the adjusted charge for high-intensity conventional neonatal care was estimated as £9901. There is now information from the trial about the chances of death before discharge. With estimates from current charge information based on the time infants spend at different levels of care, a very rough estimate of the purchase price per survivor would be £24123 for ECMO and £23450 for conventional management. Until the full cost-effectiveness study based on the trial data is completed, the charges purchasers may face for conventional and ECMO management should be taken as imperfect indicators of the relative costs of the initial hospital stay. The cost to parents will also be considered, particularly whether any time away from home, which normally occurs when an infant is referred for ECMO, is associated with additional financial burdens on the family.

Not all eligible UK infants were recruited to the trial because some neonatal units did not participate and other babies may not have been considered for the trial. We estimate that the number of infants likely to be presenting to ECMO centres after the trial will be larger, perhaps 1 in 4000 livebirths, equivalent to about 190 annually in the UK.^{19,20} The demand may be further increased if infants with less severe respiratory failure are considered for ECMO. Various indices of disease severity have been used in routine clinical practice, and the trial dataset will be explored to judge their relative merits in this setting. An oxygenation index of 40 as an entry criterion seems to represent a reasonable marker of disease severity sufficient to warrant ECMO. The rates of mortality and morbidity of these very sick infants are high, and it may be that referral should take place earlier to take into account the rate of deterioration. Some infants stabilise over a period of days, with oxygenation indices just below 40. Once all other options have been tried, there may be no advantage in delaying referral of these infants beyond 12 h.

The results of this trial cannot, however, be extrapolated to infants with oxygenation indices of nearer 25. Schumacher et al²¹ compared early ECMO (initiated at an oxygenation index between 25 and 40) with conventional ECMO management (initiated at a value ≥ 40) in a small trial and concluded that early use of ECMO does not increase hospital costs and suggests a lower morbidity for patients so treated. A larger trial would be required, however, to provide a sufficiently reliable result on which to base practice.

Although exclusion of infants with severe bleeding problems, pre-existing neurological damage, evidence of a substantial intraventricular haemorrhage, or exposure to high-pressure ventilation for more than 10 days seems prudent, indications for ECMO may extend in the future to include infants with chromosomal lesions (in particular Down's syndrome), when this seems appropriate. Also, extension to less mature infants may become possible if technical problems with the circuitry and the need for systemic heparinisation can be circumvented.

There are now five neonatal ECMO centres in the UK.

Two have developed primarily as extensions of the local cardiothoracic surgery service and three as part of integrated paediatric hospitals. For the future, centres providing neonatal ECMO will not only need close support from neonatologists, paediatric cardiologists, paediatric surgeons, and cardiac perfusion personnel, with access to active neonatal follow-up programmes and expert neonatal transport teams,²² they will also have to treat sufficient numbers of infants to maintain staff skills and competence. The number of centres that will be required will depend not only on neonatal referrals but also on the demand for paediatric and adult ECMO. During the trial, the UK ECMO centres operated within rigorous self-imposed standards. Mechanisms to ensure that this quality of care is maintained and improved are essential. Such measures should include recommendations about minimum activity levels, staffing, training, subspecialty support, and the organisation of transport.

The study has shown that large, rigorous, collaborative, randomised trials of treatment for life-threatening conditions can be done. The trial was supported by the England and Wales Department of Health and the Scottish Office Department of Health, and by hundreds of clinicians throughout the UK, and so may be considered as a model of the recent research and development ethos in the National Health Service.²³ We hope that the same spirit of cooperation will continue with the evaluation of other interventions in this setting, such as inhaled nitric oxide and high-frequency ventilation.

The trial would not have been possible without the participation of the parents of affected infants. A substudy is exploring the views of parents whose infants were in the trial, focusing on their views of the process of recruitment into the trial. The opinions and feelings of the clinicians about participation in this trial are also being surveyed. This project will act as an audit of the procedures used in this trial and will provide empirically based information that will be useful in planning of future trials.

UK Collaborative ECMO Trial Group

Writing Committee D Field, C Davis, D Elbourne, A Grant, A Johnson, D Macrae.

Steering group D Field (chairman), P Burton (up to 1993), C Davis, A Doulah (1995), D Elbourne (trial coordinator), A Grant (trial coordinator), A Greenough, P Hale, L Hamilton, A Johnson, S Kerr (1994), M Levene, M Liddell, F Lockett, R MacNab (up to 1993), D Macrae, G Pearson (1993), C Skeoch.

Advisory groups: Developmental follow-up D Elbourne, D Field, A Grant, A Johnson, M Levene, R Morley, L Mutch. *Health economics* D Elbourne, D Field, A Grant, A Johnson, L Hallam, S Howard, M Mugford, C Normand. *Pathology* J Bell, D Doyle, D Elbourne, M Evans, D Field, A Grant, A Howatson, I Jeffrey, J Keeling, P McKeever, R Risdon, W Squier, C Wright. *Respiratory* C Beardsmore, D Elbourne, D Field, A Johnson, J Stocks. *Technical* C Davis, M Elliott, R Firmin, L Hamilton. *Voluntary groups* F Lockett (coordinator).

Data-monitoring committee: R Doll (chairman), E Alberman, D Altman, F Cockburn, R Cooke.

ECMO trial clinical coordinating centre: A Doulah (1995), D Field, A Fenton, J Grant, W Hoskyns, S Kerr (1994), U MacFadyen, G Pearson (1993).

Developmental paediatricians: M Graham, C Jessen, A Schreuder.

Data-coordinating centre: H Ashurst (NPEU computing coordinator), S Ayers (programmer), D Elbourne (Director, Perinatal Trials Service), K Enock (administrative coordinator), J Fooks (programmer), C Harris (data manager), E Lukes (data entry), A Wrothford (programmer).

ECMO centres: (local medical coordinators, ECMO fellows, nursing coordinators, paediatric surgeons, pathologists): Freeman, Newcastle upon Tyne (J Hamilton, J Madar, G Derrick, L Scott, C Wright); Glenfield General, Leicester (G Pearson, S Kerr, A Doulah, H Moore, R Firmin); Great Ormond Street, London (D Macrae, C Pierce, L Tysczuk, A Goldman, B Boosfeld, E Smith, R Risdon); King's College

(A Greenough, V Noble, M King, J Desai, M Newbould); Royal Hospital for Sick Children, Glasgow (C Skeoch, E Thomas, A Tometzki, M Liddell, C Davis, A Howatson).

Referral hospitals: (local medical coordinators, nursing coordinators, paediatric surgeons, pathologists): Aberdeen Maternity (P Duffy, R Allan, G Youngson, E Gray); All Saints, Chatham (K Chan, V Sankus, B Randall); Arrowe Park, Wirral (D Manning, M Chessall, M Gillett); Ayrshire Central, Irvine (M Blair, A Hoyle, R Nairn); Bellshill Maternity, Strathclyde (J Whyte, M Gray, A Patrick); Birmingham Children's (C Ralston, H Watson, F Raafat); Birmingham City (J Bissenden, R Reeve, W Shortland-Webb); Birmingham Heartlands (M Watkinson, M Bateman, F Brook); Birmingham Maternity (M Morgan, M James, I Rushton); Bradford Royal (S Chatfield, L Francis, P Batman); Chelsea and Westminster, London (M Markiewicz, J Wood, N Madden, J Wigglesworth); Countess of Chester (N Murphy, D Freeman); Derby City General (K Dodd, M Boen, D Semeraro); Derriford, Plymouth (P Ward, T Mill, A Sherwood); Fazakerley, Liverpool (B Shaw, L McIvor, W Taylor); Forth Park, Kirkcaldy (C Steer, S Campbell, J Keeling); Freeman, Newcastle upon Tyne (G Derrick, L Scott, L Hamilton, C Wright); Glasgow Royal Maternity (C Skeoch, D Coats, A Howatson); Glenfield General, Leicester (A Doulah, H Moore, R Firmin); Great Ormond Street, London (D Macrae, D Drake, R Risdon); Hammersmith, London (R Mupanemunda, C Meredith, J Wigglesworth); Hillingdon, Middlesex (R Buchdahl, P Morris, A Davey); Homerton, London (K Costeloe, C Philips, H Rees); Hope, Manchester (M Robinson, J Bloor, J Bruce, A Kelsey); Jessop, Sheffield (R Coombs, S Hands, S Variend); John Radcliffe, Oxford (A Wilkinson, D Davidson, S Gould); Kent and Canterbury (N Martin, B Coates, A Gibson); Kettering General (I Mukhtar, G Nicholls, S Milkin); King's College, London (A Greenough, M King, J Desai, M Newbould); Leeds General (P Chetcuti, J Hetherington, J Beck, G Batcup); Leicester Royal (U MacFadyen, M Halfhead, P McKeever); Leighton, Crewe (A Thomson, S Cooke, A Nicol); Lewisham Children's, London (D Garvie, J Collas, E Dykes, C Keen); Liverpool Maternity (M Weindling, S Williams, D Lloyd, R van Velsen); Luton and Dunstable (P Sivakumar, C Toyer, D Lawrence); Mayday, Surrey (S Hart, L Ooi, I Jeffrey); Newcastle upon Tyne General (D Millington, B Watson, J Wagget, C Wright); Ninewells, Dundee (W Tarnow-Mordi, A Findlay, S Lang); Norfolk and Norwich (C Upton, H Butcher, J Brain, R Lonsdale); North Staffordshire Maternity, Stoke-on-Trent (S Spencer, K Lockyer, T French); North Tees General, Stockton (I Verber, J Dale, C Rettman); Northampton General (P Daish, M Mahony, A Molyneux); Northwick Park, Harrow (R Thomas, G Taylor, A Price); Nottingham City (A Leslie, D Fagan); Poole General (D Shortland, D Nicholas); Princess Anne, Southampton (M Hall, P Walton, D Burge, I Moore); Queen Charlotte's and Chelsea, London (R Mupanemunda, S Day, J Wigglesworth); Queen Elizabeth, London (D Drake, J Ashton, R Risdon); Queen Mother's Maternity, Glasgow (B Holland, J Scott, A Patrick); Queen's Medical Centre, Nottingham (J Grant, A Leslie, D Fagan); Queen's Park, Blackburn (J Benson, D Hatton, R Prescott); Raigmore, Inverness (C Galloway, R Finlayson, J MacPhie); Royal Berkshire, Reading (J Smyth, M Muncaster, L Horton); Royal Devon and Exeter (M Quinn, R Shanks, P Anthony); Royal Hospital for Sick Children, Edinburgh (G MacKinlay, P Skinner, J Keeling); Royal Hospital for Sick Children, Glasgow (C Skeoch, M Liddell, C Davis, A Howatson); Royal Liverpool Children's (J Ratcliffe, D Lloyd); Royal Shrewsbury Maternity (R Welch, S Ellis, A Fraser); Royal United, Bath (P Rudd, J Jackson, C Meehan); Royal Victoria, Newcastle upon Tyne (D Milligan, S Davies, M Barrett, C Wright); Rush Green, Romford (J Rawal, L Yong, A Tagizadeh); Sheffield Children's (J Dickson, C McLoughlin, S Variend); Simpson Memorial Maternity, Edinburgh (I Laing, M Howatt, J Keeling); South Cleveland, Middlesbrough (S Sinha, J Gavey, U Earl); Southampton General (D Burge); Southern General, Glasgow (M White, R Nicholson, A Patrick); St George's, London (S Calvert, L Weir, K Holmes, I Jeffrey); St Helier, Carshalton (K Haque, A-M Hayes, W Landells); St Mary's, London (S Bignall, D Summers, N Maddon, M Wilkin); St Mary's, Portsmouth (M Ashton); St Peter's, Chertsey (J Bowyer, L Moore, M Hall); St Thomas', London (R Thwaites, M Jackson, H Ward, N Fagg); Royal London (S Kempley, V Mendham, V Wright, C Berry); Royal Wolverhampton Hospitals NHS Trust (J Anderson, C Price, J Bridger); University Hospital of Wales, Cardiff (M Drayton, M Reed, J Lari, G Vujanic); Walsgrave, Coventry (A Coe, M Newsome); Whittington, London (E Broadhurst, J Greenwood, D Brown); Wycombe General, High Wycombe (C Cheetham, J MacDonald, M Turner).

Other participating hospitals: Alexandra, Redditch (S Perera); Antrim Area (J Lim); Ashington (J Oliver); Basildon (S Ware); Bishop Auckland (W Lamb); City General, Carlisle (C Stuart); Craigavon Area (M O'Connor); Crawley (I Lewis); Derbyshire Children's (N Ruggins); Eastbourne District General (S Birks); Falkirk and District Royal (D Hannona); George Eliot, Nuneaton (A Comley); Guy's, London (I Murdoch); Hemel Hempstead General (H El-Naggar); Huddersfield Royal (A Short); King's Mill, Sutton-in-Ashfield (A Worsley); Lister, Stevenage (J Reiser); Nevill Hall, Abergavenny (T Williams); Newcastle General (S Srinivas); Newham General (J Allgrove); North Tyneside

General, North Shields (W Houlshby); Pindersfields General, Wakefield (S Jones); Queen Elizabeth Hospital, Gateshead (M Higgs); Queen Elizabeth II, Welwyn Garden City (A Tybulewicz); Rotherham District General (P MacFarlane); Royal Gwent, Newport (S Ferguson); Royal Hampshire, Winchester (D Schapira); Royal Sussex County, Brighton (P Seddon); St Paul's Cheltenham (A Day); Selly Oak, Birmingham (M Hocking); Singleton, Swansea (J Matthes); Stoke Mandeville, Aylesbury (C Noone); Royal Free, London (V van Someren); University College, London (J Hawdon); Watford General (G Supramaniam); William Harvey, Ashford (C Porter).

Health visitors: E Abbey, E Adu, J Allsopp, S Atkinson, T Barnham, D Bayliss, W Beeley, J Beeny, G Bennett, P Bone, A Bower, S Brown, M Buchanan, A Burling, J Burrows, L Byrne, S Calabrese, D Campbell, P Ceres, R Chana, M Chesney, S Clare, P Cogan, M Cook, K Coote, M Cornforth, K Coupland, R Cullinan, D Davies, D Davison, M Dow, M Drake, B Durkin, C Edwards, J Falconer, B Ferguson, F Forrest, M Gallagher, I Grant, P Gregson, U Gunn, T Hall, V Hames, M Hamilton, J Hart, C Hazlehurst, A Henry, F Hill, C Hogan, M Hubbard, T Jackson, I Keating, E Kelly, J Lawrie, A Lee, E Leighton, R Lindsay, Z Linfield, P Long, M Love, A Mackness, P Martin, A McClelland, A Mills, F Monck, P Neal, C Norgate, R Pavlou, S Peck, M Phillips, E Pollock, J Rankin, D Rawson, K Rehling, L Reijnders, S Ryan, F Sandilands, S Savage, M Savill, A Scott, J Self, A Shaw-Flach, A Sibson, K Simpson, M Sklar, H Spencer, S Stones, L Stonestreet, R Streete, M Taylor, M Urquhart, S Van Der Vliet, J Waddingham, A Walker, N Wallace, A Weir, M West, S West, W Williamson, M Woodhouse, P Wykes, A Young.

Acknowledgments

We thank the 185 babies and their parents who took part in the trial, and the hundreds of doctors and nurses and their colleagues in the participating centres; the Clinical Trial Service Unit, Oxford (J Crowther, J Heineman, S Knight, A Naughten, A Radley, S Wenzel) for the 24 h central randomisation service; colleagues at the NPEU and elsewhere for their helpful advice and encouragement (particularly Edmund Hey and Iain Chalmers); and Ellen Bifano and Robert Bartlett for sharing unpublished data with us. Throughout the trial, we were fortunate to have the opportunity to listen to the views of and to have warm support and helpful advice from many voluntary organisations concerned with maternity services. The study was supported by the England and Wales Department of Health, and the Chief Scientist's Office, Scottish Office Department of Health. The Wellcome Trust provided some funding for the study on participants' views and for the respiratory follow-up. The Perinatal Trials Service and the National Perinatal Epidemiology Unit are funded by the Department of Health.

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